# 2013 Team Answers

T-1. 12

T-2. 1525

T-3. 272

T-4. 2019

T-5. 50

T-6. $-\frac{1}{2}$

T-7. 14

T-8. $-21$

T-9. $\frac{5\sqrt{39}}{3}$

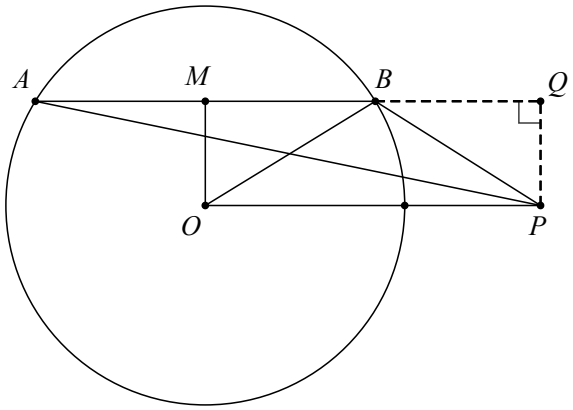
T-10. 140

# 2013 Team Solutions

T-1. In order for $1584 \cdot x$ to be a perfect cube, all of its prime factors must be raised to powers divisible by 3. Because $1584 = 2^4 \cdot 3^2 \cdot 11$, $x$ must be of the form $2^{3k+2} \cdot 3^{3m+1} \cdot 11^{3n+2} \cdot r^3$, for nonnegative integers $k, m, n, r$, $r > 0$. Thus the least positive value of $x$ is $2^2 \cdot 3 \cdot 11^2 = 1452$. But in order for $xy$ to be a positive multiple of 1584, $xy$ must be of the form $2^a \cdot 3^b \cdot 11^c \cdot d$, where $a \geq 4$, $b \geq 2$, $c \geq 1$, and $d \geq 1$. Thus $y$ must equal $2^2 \cdot 3^1 = \mathbf{12}$.

T-2. Let $a$ be the price of one apple and $p$ be the price of one peach, in cents. The first transaction shows that $500 < 5a + 5p < 1000$, hence $100 < a + p < 200$. The second transaction shows that $1000 < 2a + 12p < 2000$, so $500 < a + 6p < 1000$. Subtracting the inequalities yields $300 < 5p < 900$, so $60 < p < 180$. Therefore the price of 25 peaches is at least $25 \cdot 61 = \mathbf{1525}$ cents.

T-3. Extend $\overline{AB}$ to point $Q$ such that $\overline{PQ} \perp \overline{AQ}$ as shown, and let $M$ be the midpoint of $\overline{AB}$. (The problem does not specify whether $A$ or $B$ is nearer $P$, but $B$ can be assumed to be nearer $P$ without loss of generality.)




Then $OP = 10$, $PQ = OM = 2$, and $OB = 6$. Thus $MB = \sqrt{6^2 - 2^2} = 4\sqrt{2}$. Because $QM = OP = 10$, it follows that $QB = 10 - 4\sqrt{2}$ and $QA = 10 + 4\sqrt{2}$. So

$$\begin{aligned} PA^2 + PB^2 &= (QA^2 + QP^2) + (QB^2 + QP^2) \\ &= (10 + 4\sqrt{2})^2 + 2^2 + (10 - 4\sqrt{2})^2 + 2^2 \\ &= \mathbf{272}. \end{aligned}$$

T-4. If $a + b \geq 2014$, then at least one of $a, b$ must be greater than 1006. The palindromes greater than 1006 but less than 2014 are, in descending order, 2002, 1991, 1881, ..., 1111. Let $a$

represent the larger of the two palindromes. Then for $n = 2014$, $a = 2002$ is impossible, because $2014 - 2002 = 12$. Any value of $a$ between 1111 and 2000 ends in 1, so if $a + b = 2014$, $b$ ends in 3, and because $b < 1000$, it follows that $303 \le b \le 393$. Subtracting 303 from 2014 yields 1711, and so $a \le 1711$. Thus $a = 1661$ and $b = 353$. A similar analysis shows the following results:

$$\begin{aligned} 2015 &= 1551 + 464; \\ 2016 &= 1441 + 575; \\ 2017 &= 1331 + 686; \text{ and} \\ 2018 &= 1221 + 797. \end{aligned}$$

But **2019** cannot be expressed as the sum of two palindromes: $b$ would have to end in 8, so $b = 808 + 10d$ for some digit $d$. Then $2019 - 898 \le a \le 2019 - 808$, hence $1121 \le a \le 1211$, and there is no palindrome in that interval.

T-5. First consider the problem with $x, y, z$ positive real numbers. If $xy + z = 160$ and $z$ is constant, then $y = \frac{160-z}{x}$, yielding $x + yz = x + \frac{z(160-z)}{x}$. For $a, x > 0$, the quantity $x + \frac{a}{x}$ is minimized when $x = \sqrt{a}$ (proof: use the Arithmetic-Geometric Mean Inequality $\frac{A+B}{2} \ge \sqrt{AB}$ with $A = x$ and $B = \frac{a}{x}$); in this case, $x + \frac{a}{x} = 2\sqrt{a}$. Thus $x + yz \ge 2\sqrt{z(160 - z)}$. Considered as a function of $z$, this lower bound is increasing for $z < 80$.

These results suggest the following strategy: begin with small values of $z$, and find a factorization of $160 - z$ such that $x$ is close to $\sqrt{z(160 - z)}$. (Equivalently, such that $\frac{x}{y}$ is close to $z$.) The chart below contains the triples $(x, y, z)$ with the smallest values of $x + yz$, conditional upon $z$.

| $z$ | $(x, y, z)$ | $x + yz$ |
|---|---|---|
| 1 | $(53, 3, 1)$ | 56 |
| 2 | $(79, 2, 2)$ | 83 |
| 3 | $(157, 1, 3)$ | 160 |
| 4 | $(26, 6, 4)$ | 50 |
| 5 | $(31, 5, 5)$ | 56 |
| 6 | $(22, 7, 6)$ | 64 |

Because $x + yz \ge 2\sqrt{z(160 - z)}$, it follows that $x + yz \ge 64$ for $6 \le z \le 80$. And because $x + yz > 80$ for $z \ge 80$, the minimal value of $x + yz$ is **50**.

**Note:** In fact, 160 is the smallest positive integer such that $xy + z = N$ requires $z \ge 4$ in the minimal solution for $x + yz$. The smallest values of $N$ for which $z = 2, \ldots, 12$, are 8, 48, 160, 720, 790, 1690, 4572, 13815, 22031, 22032, 79965.

T-6. The identity $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ can be rewritten into the power-reducing identity

$$\cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta.$$

Thus if $D$ is the desired sum,

$$\begin{aligned} D &= \cos^3\frac{2\pi}{7} + \cos^3\frac{4\pi}{7} + \cos^3\frac{8\pi}{7} \\ &= \frac{1}{4}\left(\cos\frac{6\pi}{7} + \cos\frac{12\pi}{7} + \cos\frac{24\pi}{7}\right) + \frac{3}{4}\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}\right). \end{aligned}$$

Observe that $\cos\frac{24\pi}{7} = \cos\frac{10\pi}{7}$, so

$$D = \frac{1}{4}\left(\cos\frac{6\pi}{7} + \cos\frac{12\pi}{7} + \cos\frac{10\pi}{7}\right) + \frac{3}{4}\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}\right).$$

Notice also that $\cos\theta = \cos(2\pi - \theta)$ implies $\cos\frac{12\pi}{7} = \cos\frac{2\pi}{7}$, $\cos\frac{10\pi}{7} = \cos\frac{4\pi}{7}$, and $\cos\frac{8\pi}{7} = \cos\frac{6\pi}{7}$. Rewriting $D$ using the least positive equivalent angles yields

$$\begin{aligned} D &= \frac{1}{4}\left(\cos\frac{6\pi}{7} + \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7}\right) + \frac{3}{4}\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}\right) \\ &= \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}. \end{aligned}$$

To evaluate this sum, use the identity $\cos\theta = \cos(2\pi - \theta)$ again to write

$$2D = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{8\pi}{7} + \cos\frac{10\pi}{7} + \cos\frac{12\pi}{7}.$$

If $\alpha = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$, notice that the right side of the equation above is simply the real part of the sum $\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6$. Because $\alpha^n$ is a solution to the equation $z^7 = 1$ for $n = 0, 1, \ldots, 6$, the sum $1 + \alpha + \alpha^2 + \cdots + \alpha^6$ equals 0. Hence $\alpha + \alpha^2 + \cdots + \alpha^6 = -1$ and $D = \mathbf{-1/2}$.

**Alternate Solution:** Construct a cubic polynomial in $x$ for which $\cos\frac{2\pi}{7}$, $\cos\frac{4\pi}{7}$, and $\cos\frac{8\pi}{7}$ are zeros; then the sum of their cubes can be found using techniques from the theory of equations. In particular, suppose the three cosines are zeros of $x^3 + bx^2 + cx + d$. Then

$$\begin{aligned} b &= -\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}\right), \\ c &= \cos\frac{2\pi}{7}\cos\frac{4\pi}{7} + \cos\frac{2\pi}{7}\cos\frac{8\pi}{7} + \cos\frac{4\pi}{7}\cos\frac{8\pi}{7}, \text{ and} \\ d &= -\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{8\pi}{7}. \end{aligned}$$

Use complex seventh roots of unity (as in the previous solution) to find $b = 1/2$. To find $c$, use the product-to-sum formula $2\cos A\cos B = \cos(A + B) + \cos(A - B)$ three times:

$$\begin{aligned} 2c &= \left(\cos\frac{6\pi}{7} + \cos\frac{2\pi}{7}\right) + \left(\cos\frac{10\pi}{7} + \cos\frac{6\pi}{7}\right) + \left(\cos\frac{4\pi}{7} + \cos\frac{12\pi}{7}\right) \\ &= \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{8\pi}{7} + \cos\frac{10\pi}{7} + \cos\frac{12\pi}{7} \text{ [because } \cos\theta = \cos(2\pi - \theta)] \\ &= -1. \end{aligned}$$

Thus $c = -1/2$.

To compute $d$, multiply both sides by $\sin\frac{2\pi}{7}$ and use the identity $2\sin\theta\cos\theta = \sin 2\theta$:

$$\begin{aligned} d\sin\frac{2\pi}{7} &= -\sin\frac{2\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{8\pi}{7} \\ &= -\frac{1}{2}\sin\frac{4\pi}{7}\cos\frac{4\pi}{7}\cos\frac{8\pi}{7} \\ &= -\frac{1}{4}\sin\frac{8\pi}{7}\cos\frac{8\pi}{7} \\ &= -\frac{1}{8}\sin\frac{16\pi}{7}. \end{aligned}$$

Because $\sin\frac{16\pi}{7} = \sin\frac{2\pi}{7}$, the factors on both sides cancel, leaving

$$d = -1/8.$$

Thus $\cos\frac{2\pi}{7}$, $\cos\frac{4\pi}{7}$, and $\cos\frac{8\pi}{7}$ are roots of $x^3 + \frac{1}{2}x^2 - \frac{1}{2}x - \frac{1}{8}$; so each value also satisfies the equation $x^3 = -\frac{1}{2}x^2 + \frac{1}{2}x + \frac{1}{8}$. Hence the desired sum can be rewritten as

$$\begin{aligned} \cos^3\frac{2\pi}{7} + \cos^3\frac{4\pi}{7} + \cos^3\frac{8\pi}{7} &= -\frac{1}{2}\left(\cos^2\frac{2\pi}{7} + \cos^2\frac{4\pi}{7} + \cos^2\frac{8\pi}{7}\right) \\ &+ \frac{1}{2}\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7}\right) + \frac{3}{8}. \end{aligned}$$

Previous work has already established that $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7} = -1/2$, so it remains to compute $\cos^2\frac{2\pi}{7} + \cos^2\frac{4\pi}{7} + \cos^2\frac{8\pi}{7}$. The identity $A^2+B^2+C^2 = (A+B+C)^2 - 2(AB+BC+AC)$ allows the use of previous results: $\cos^2\frac{2\pi}{7} + \cos^2\frac{4\pi}{7} + \cos^2\frac{8\pi}{7} = (-1/2)^2 - 2(-1/2) = 5/4$. Thus

$$\cos^3\frac{2\pi}{7} + \cos^3\frac{4\pi}{7} + \cos^3\frac{8\pi}{7} = -\frac{1}{2}\left(\frac{5}{4}\right) + \frac{1}{2}\left(-\frac{1}{2}\right) + \frac{3}{8} = \boxed{-\frac{1}{2}}.$$

T-7. Note that in right triangle $ABC$ with right angle $C$, the inradius $r$ is equal to $\frac{a+b-c}{2}$, where $a = BC$, $b = AC$, and $c = AB$, because the inradius equals the distance from the vertex of the right angle $C$ to (either) point of tangency along $\overline{AC}$ or $\overline{BC}$. Thus the sum of the inradii of triangles $AXP, CXS, CYR$, and $BYQ$ is equal to one-half the difference between the sum of the lengths of the legs of these triangles and the sum of the lengths of the hypotenuses of these triangles. Let $t$ be the side length of square $PQRS$. Then the sum of the lengths of the legs of triangles $AXP, CXS, CYR$, and $BYQ$ is

$$\begin{aligned} & AP + PX + XS + SC + CR + RY + YQ + QB \\ = \; & AP + PS + SR + RQ + QB \\ = \; & AP + t + t + t + QB \\ = \; & AB - PQ + 3t \\ = \; & c - t + 3t \\ = \; & c + 2t. \end{aligned}$$

The sum of the lengths of the hypotenuses of triangles $AXP, CXS, CYR$, and $BYQ$ is $AX + XC + CY + YB = AC + CB = b + a$. Hence the sum of the inradii of triangles $AXP, CXS, CYR$, and $BYQ$ is $\frac{c+2t-(a+b)}{2} = t - r$. Thus the desired sum equals $\sqrt{576} - 10 = 24 - 10 = \mathbf{14}$.

T-8. Because the quantity on the left side is the difference of two integers, $x/7$ must be an integer, hence $x$ is an integer (in fact a multiple of 7). Because the denominators on the left side are 2 and 3, it is convenient to write $x = 6q + r$, where $0 \le r \le 5$, so that $\lfloor x/2 \rfloor = 3q + \lfloor r/2 \rfloor$ and $\lfloor x/3 \rfloor = 2q + \lfloor r/3 \rfloor$. Then for $r = 0, 1, \ldots, 5$ these expressions can be simplified as shown in the table below.

| $r$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $\lfloor \frac{x}{2} \rfloor$ | $3q$ | $3q$ | $3q+1$ | $3q+1$ | $3q+2$ | $3q+2$ |
| $\lfloor \frac{x}{3} \rfloor$ | $2q$ | $2q$ | $2q$ | $2q+1$ | $2q+1$ | $2q+1$ |
| $\lfloor \frac{x}{2} \rfloor - \lfloor \frac{x}{3} \rfloor$ | $q$ | $q$ | $q+1$ | $q$ | $q+1$ | $q+1$ |

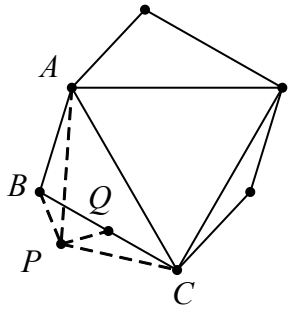
Now proceed by cases:

$r = 0$: Then $q = x/6$. But from the statement of the problem, $q = x/7$, so $x = 0$.
$r = 1$: Then $q = (x-1)/6 = x/7 \Rightarrow x = 7$.
$r = 2$: Then $q = (x-2)/6$ and $q + 1 = x/7$, so $(x+4)/6 = x/7$, and $x = -28$.
$r = 3$: Then $q = (x-3)/6$ and $q = x/7$, so $x = 21$.
$r = 4$: Then $q = (x-4)/6$ and $q + 1 = x/7$, so $(x+2)/6 = x/7$, and $x = -14$.
$r = 5$: Then $q = (x-5)/6$ and $q + 1 = x/7$, so $(x+1)/6 = x/7$, and $x = -7$.

The sum of these values is $0 + 7 + -28 + 21 + -14 + -7 = \mathbf{-21}$.

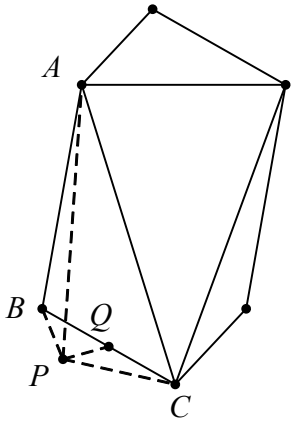
T-9. First consider a regular octahedron of side length 1. To compute its volume, divide it into two square-based pyramids with edges of length 1. Such a pyramid has slant height $\frac{\sqrt{3}}{2}$ and height $\sqrt{(\frac{\sqrt{3}}{2})^2 - (\frac{1}{2})^2} = \sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2}$, so its volume is $\frac{1}{3} \cdot 1^2 \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{6}$. Thus the octahedron has volume twice that, or $\frac{\sqrt{2}}{3}$. The result of folding the net shown is actually the image of a regular octahedron after being stretched along an axis perpendicular to one face by a factor of $r$. Because the octahedron is only being stretched in one dimension, the volume changes by the same factor $r$. So the problem reduces to computing the factor $r$ and the edge length of the resulting octahedron.

For convenience, imagine that one face of the octahedron rests on a plane. Seen from above the plane, the octahedron appears as shown below.




Let $P$ be the projection of $A$ onto the plane on which the octahedron rests, and let $Q$ be the foot of the perpendicular from $P$ to $\overline{BC}$. Then $PQ = R_C - R_I$, where $R_C$ is the circumradius and $R_I$ the inradius of the equilateral triangle. Thus $PQ = \frac{2}{3}\left(\frac{\sqrt{3}}{2}\right) - \frac{1}{3}\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{6}$. Then $BP^2 = PQ^2 + BQ^2 = \frac{3}{36} + \frac{1}{4} = \frac{1}{3}$, so $AP^2 = AB^2 - BP^2 = \frac{2}{3}$, and $AP = \frac{\sqrt{6}}{3}$.

Now let a vertical stretch take place along an axis parallel to $\overleftrightarrow{AP}$. If the scale factor is $r$, then $AP = \frac{r\sqrt{6}}{3}$, and because the stretch occurs on an axis perpendicular to $\overline{BP}$, the length $BP$ is unchanged, as can be seen below.




Thus $AB^2 = \frac{6r^2}{9} + \frac{1}{3} = \frac{6r^2+3}{9}$. It remains to compute $r$. But $r$ is simply the ratio of the new volume to the old volume:

$$r = \frac{6}{\frac{\sqrt{2}}{3}} = \frac{18}{\sqrt{2}} = 9\sqrt{2}.$$

Thus $AB^2 = \frac{6(9\sqrt{2})^2+3}{9} = \frac{975}{9} = \frac{325}{3}$, and $AB = \mathbf{\frac{5\sqrt{39}}{3}}$.

T-10. If $f$ is simply a permutation of $S$, then $\{s_n\}$ is periodic. To understand why, consider a smaller set $T = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. If $f : [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] \to [2, 3, 4, 5, 1, 7, 8, 6, 9, 10]$, then $f$ has one cycle of period 5 and one cycle of period 3, so the period of $f$ is 15. However,

$$f(1) + f(2) + f(3) + f(4) + f(5) + f(6) + f(7) + f(8) + f(9) + f(10) =$$

$$2+3+4+5+1+7+8+6+9+10=55,$$

because $f$ just rearranges the order of the summands. So $s_1 = s_0$, and for all $n$, $s_n = s_{n+1}$; in short, the period of $\{s_n\}$ is just 1.

In order for $\{s_n\}$ to have a period greater than 1, $f$ must be many-to-one, so that some values occur more than once (and some values do not occur at all) in the sum $f(1)+f(2)+\cdots+f(10)$ (or, in the original problem, $f(1)+f(2)+\cdots+f(20)$). For example, consider the function $f_2$ below:

$$f_2 : [1,2,3,4,5,6,7,8,9,10] \to [2,3,4,5,1,10,9,10,7,3].$$

Note that $s_1 = 2+3+4+5+1+10+9+10+7+3 \neq 55$, so $\{s_n\}$ is not immediately periodic. But $\{s_n\}$ is eventually periodic, as the following argument shows. The function $f_2$ has two cycles: $1 \to 2 \to 3 \to 4 \to 5 \to 1$, and $7 \to 9 \to 7$. There are also two paths that meet up with the first cycle: $6 \to 10 \to 3 \to \cdots$ and $8 \to 10 \to 3 \to \cdots$. Thus for all $k$ in $T$, $f_2(f_2(k))$ is an element of one of these two extended cycles. Thus $\{s_n\}$ eventually becomes periodic.

The criterion that the function be many-to-one is necessary, but not sufficient, for $\{s_n\}$ to have period greater than 1. To see why, consider the function $g : [1,2,3,4,5,6,7,8,9,10] \to [2,3,4,5,6,1,8,7,8,7]$. This function is many-to-one, and contains two cycles, $1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 1$ and $7 \to 8 \to 7$. But because $g(9) = 8$ and $g(10) = 7$, the sum $s_1 = 2+3+4+5+6+1+8+7+8+7$, while $s_2 = 3+4+5+6+1+2+7+8+7+8$. In fact, for $n > 1$, $s_{n+1} = s_n$, because applying $f$ only permutes the 6-cycle and switches the two 7's and two 8's. That is, in the list $\underbrace{(g \circ \cdots \circ g)}_{n}(1), \ldots, \underbrace{(g \circ \cdots \circ g)}_{n}(10)$, the values 7 and 8 both show up exactly twice. This cycle is *balanced*: each of its elements shows up the same number of times for all $n$ in the list $\underbrace{(g \circ \cdots \circ g)}_{n}(1), \ldots, \underbrace{(g \circ \cdots \circ g)}_{n}(10)$, for all $n$ after a certain point. The conclusion is that not all many-to-one functions produce unbalanced cycles.

There are two ways a function $g$ can produce balanced cycles. First, the cycles can be self-contained, so no element outside of the cycle is ever absorbed into the cycle, as happens with the 6-cycle in the example above. Alternatively, the outside elements that are absorbed into a cycle can all arrive at different points of the cycle, so that each element of the cycle occurs equally often in each iteration of $g$. In the example above, the values $g(9) = 7$ and $g(10) = 8$ balance the $7 \to 8 \to 7$ cycle. On the other hand, in the function $f_2$ above, $f(f(6)) = f(f(8)) = f(f(1)) = 3$, making the large cycle unbalanced: in $s_2$, the value 3 appears three times in $s_2$, but the value 2 only appears once in $s_2$.

The foregoing shows that only *unbalanced* cycles can affect the periodicity of $\{s_n\}$. Because each element of a balanced cycle occurs equally often in each iteration, the period of that component of the sum $s_n$ attributed to the cycle is simply 1. (The case where $f$ is a permutation of $S$ is simply a special case of this result.) In the above example, the large cycle is

unbalanced. Note the following results under $f_2$.

| $n$ | $\overbrace{(f_2 \circ \cdots \circ f_2)}^{n}(T)$ | $s_n$ |
|---|---|---|
| 1 | $[2,3,4,5,1,10,9,10,7,3]$ | 54 |
| 2 | $[3,4,5,1,2,3,7,3,9,4]$ | 41 |
| 3 | $[4,5,1,2,3,4,9,4,7,5]$ | 40 |
| 4 | $[5,1,2,3,4,5,7,5,9,1]$ | 42 |
| 5 | $[1,2,3,4,5,1,9,1,7,2]$ | 35 |
| 6 | $[2,3,4,5,1,2,7,2,9,3]$ | 38 |
| 7 | $[3,4,5,1,2,3,9,3,7,4]$ | 41 |
| 8 | $[4,5,1,2,3,4,7,4,9,5]$ | 40 |
| 9 | $[5,1,2,3,4,5,9,5,7,1]$ | 42 |

The period of $\{s_n\}$ for $f_2$ is 5, the period of the unbalanced cycle.

The interested reader may inquire whether *all* unbalanced cycles affect the periodicity of $\{s_n\}$; we encourage those readers to explore the matter independently. For the purposes of solving this problem, it is sufficient to note that unbalanced cycles *can* affect $\{s_n\}$'s periodicity.

Finally, note that an unbalanced $k$-cycle actually requires at least $k+1$ elements: $k$ to form the cycle, plus at least 1 to be absorbed into the cycle and cause the imbalance. For the original set $S$, one way to create such an imbalance would be to have $f(20) = f(1) = 2, f(2) = 3, f(3) = 4, \ldots, f(19) = 1$. This arrangement creates an unbalanced cycle of length 19. But breaking up into smaller unbalanced cycles makes it possible to increase the period of $\{s_n\}$ even more, because then in most cases the period is the least common multiple of the periods of the unbalanced cycles. For example, $f : [1,2,3,\ldots,20] = [2,3,4,5,6,7,8,9,1,1,12,13,14,15,16,17,18,11,11,11]$ has an unbalanced cycle of length 9 and an unbalanced cycle of length 8, giving $\{s_n\}$ a period of 72.

So the goal is to maximize $\text{lcm}\{k_1, k_2, \ldots, k_m\}$ such that $k_1 + k_2 + \cdots + k_m + m \leq 20$. With $m = 2$, the maximal period is 72, achieved with $k_1 = 9$ and $k_2 = 8$. With $m = 3$, $k_1 + k_2 + k_3 \leq 17$, but $\text{lcm}\{7,6,4\} = 84 < \text{lcm}\{7,5,4\} = 140$. This last result can be obtained with unbalanced cycles of length 4, 5, and 7, with the remaining four points entering the three cycles (or with one point forming a balanced cycle of length 1, i.e., a fixed point). Choosing larger values of $m$ decreases the values of $k$ so far that they no longer form long cycles: when $m = 4$, $k_1 + k_2 + k_3 + k_4 \leq 16$, and even if $k_4 = 2$, $k_3 = 3$, and $k_2 = 5$, for a period of 30, the largest possible value of $k_1 = 6$, which does not alter the period. (Even $k_1 = 7$, $k_2 = 5$, and $k_3 = k_4 = 2$ only yields a period of 70.) Thus the maximum period of $s_n$ is **140**. One such function $f$ is given below.

| $n$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $f(n)$ | 2 | 3 | 4 | 1 | 1 | 7 | 8 | 9 | 10 | 6 | 6 | 13 | 14 | 15 | 16 | 17 | 18 | 12 | 12 | 20 |